Data Handling: Import, Cleaning and Visualisation

Exercise/Workshop 2: Computer Code and Data Storage

Ulrich Matter

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Prerequisites

- Install the latest version of the text editor Atom here: https://atom.io/.

Exercises

Exercise A: Loading text files in RStudio

When working with data, a first important step is to load your data, in our case in RStudio. As you know, data can come in different formats.

1. Along with the exercise instructios on GitHub classroom, you will find two datasets, `simple_data.txt` and `simple_data.csv`. Download them from GitHub and upload them into the `r_course/data` folder that we created in the last exercise session.
2. Check out the files `simple_data.txt` and `simple_data.csv` by opening them (outside of RStudio) in a text editor, such as Atom. Do they look different from each other?
3. Go back to RStudio and load the data in .txt format as follows:

```r
df1 <- read.table("data/simple_data.txt", header = FALSE)
```

Type `df1` or `print(df1)` into the console. What do you see?

4. Next, turn to the file in .csv format. Run the following code to load it:

```r
df2 <- read.table("data/simple_data.csv", header = FALSE, sep=",")
```

Again, type `df2` or `print(df2)` into the console. Now, what do you see? Compare this output to the output from `print(df1)`.

5. Generate a text file for the first part of the autograded assignment on GitHub classroom: create a text file containing the value in the third column of the second row of `df1`. Hint: you can obtain values from dataframes by indexing like this: `df[i,j]`, where i and j indicate row and column numbers, respectively. Save the text file as "df1_value.txt" in your `r_course/data` folder.

Exercise B: Downloading and importing data on COVID cases

In the following, we are going to download and import data on COVID-19 cases around the world. We will reuse this data in later exercise sessions.

1. Using a browser of your choice, navigate to this page by the EU Open Data Portal: https://data.europa.eu/data/datasets/covid-19-coronavirus-data-daily-up-to-14-december-2020?locale=en.

2. Scroll down to the Section 'Distributions' and download the file 'COVID-19 cases worldwide - daily' in .csv format into your `r_course/data` folder.

3. Based on Exercise A, Part 4, how would you open this dataset in RStudio? Hint: In contrast to the files in Exercise A, this file comes with a header row. Complete the following code accordingly:

```r
df_covid <- read.table(...)
```

4. Once you have loaded the data into RStudio, you can inspect the first (or last) couple of rows using `head(df_covid)` (or `tail(df_covid)`).

5. Again, create a text file for upload on GitHub classroom. It should contain the number of cases (variable `cases`) on 26 March 2020 (variable `dateRep`) in Zimbabwe (variable `countriesAndTerritories`). Hint: this observation is in the 61896th row. Save you text file as "covid.txt" in your `r_course/data` folder.

Exercise C: Writing/Exploring computer code: a simple website

In this exercise, we will implement a very simple website ourselves (in HTML).

1. Open a new text file in RStudio: `File/New File/Html File`, store it in your current working directory (set the working directory accordingly beforehand) as `my_website.html`.

2. Copy/paste the follwing HTML-code to the empty text-file and save it.

```html
<!DOCTYPE html>

<html>
    <head>
    </head>
    <body>
        <h2> hello, world </h2>
        <p>Welcome to my website! How are you today?</p>
    </body>
</html>
```

3. Click on the file in the RStudio file browser and select `View in Web Browser`. The webpage should open in your default web browser. Inspect the webpage's content and compare it with the raw source code. Figure out which part of the source code is resulting in which part displayed in browser, by changing the text in the source code (change the text between the html-tags in the source code, save the file, reopen it in the web browser, etc.).

4. Visit the following documentation on the `<title>`-element: https://developer.mozilla.org/en-US/docs/Web/HTML/Element/title . Based on the description and the examples shown there, add the title `My Website` to your webpage by modifying/extending the source code accordingly. Verify your solution by checking whether the page's title is displayed in the browser tab/window.

Advanced Exercises (exam-relevant)

Exercise D: Downloading and importing data on COVID cases (XML format)

1. Go back to the EU Open Data Portal (see Exercise B). Download the same dataset again, but this time, download the .xml version. Again, save it in your `r_course/data` folder. (Loading the)
2. Load the data in .xml format. Hint: The package `XML` might be of use. You can install new packages as follows: `install.packages("package_name")`.
3. Once you have loaded the .xml file in Part 2, how can you convert it to a R dataframe?

Exercise E: Read raw text data

This exercise guides you through how we can import raw text data into R. Suppose you are writing your BA Thesis on the topic "The Effect of Climate Change on Consumption Behavior in Switzerland". For your empirical analysis you need inter alia standardized data on temperature anomalies in Switzerland (as a proxy for the salience of climate change). You identify NASA's Earth System Data Explorer as a promising data source and aim to download and work with this data in R. The exercise guide's you exemplary thourgh how this can be done.

Hint: This exercise is an adapted version of the tutorial in Chapter 1 of Paul Murell's book (see syllabus). Read the chapter for an additional introduction to data stored in text files.

1. Go to NASA's Earth System Data Explorer: https://mynasadata.larc.nasa.gov/EarthSystemLAS/UI.vm.
2. Click on *Data Set* in the upper left corner and navigate through `Atmosphere/Featured Phenomena/Changing Air Temperatures/Temperatures` and select `Monthly Surface Air Temperature Anomaly`.
3. On the bottom left part of the page, under `Line Graphs`, select `Time`.
4. Enter the coordinates 47.25 N and 9.22 W (St. Gallen) and click `Update Plot` in the upper left corner. The page shows a plot of the monthly surface air temperature anomaly time series.
5. Click on `Download Data ...`. In the new tab select `ASCII` as a data format. Click to save.

The downloaded ASCII-file containing the raw data should open in a new tab. Inspect the data and save (`Save page as...`) it as `temp_anomaly.txt` in your `r_course/data` folder.

Now we can read the data into R:

```r
# install the readr package if not yet installed: install.packages("readr")
library(readr)
# read the data
ta <- read_tsv("data/temp_anomaly.txt", skip = 9, col_names = c("month", "temp_anomaly"))
```

```r
# inspect the first rows
head(ta)
```

- What is the purpose of the `skip` argument and why is it set to `9`?
- Why do we use `read_tsv()` in this situation and not `read_csv()`?